CLOVE

CLOVE: Continual Learning for Open-World Visual Evaluation in Agriculture

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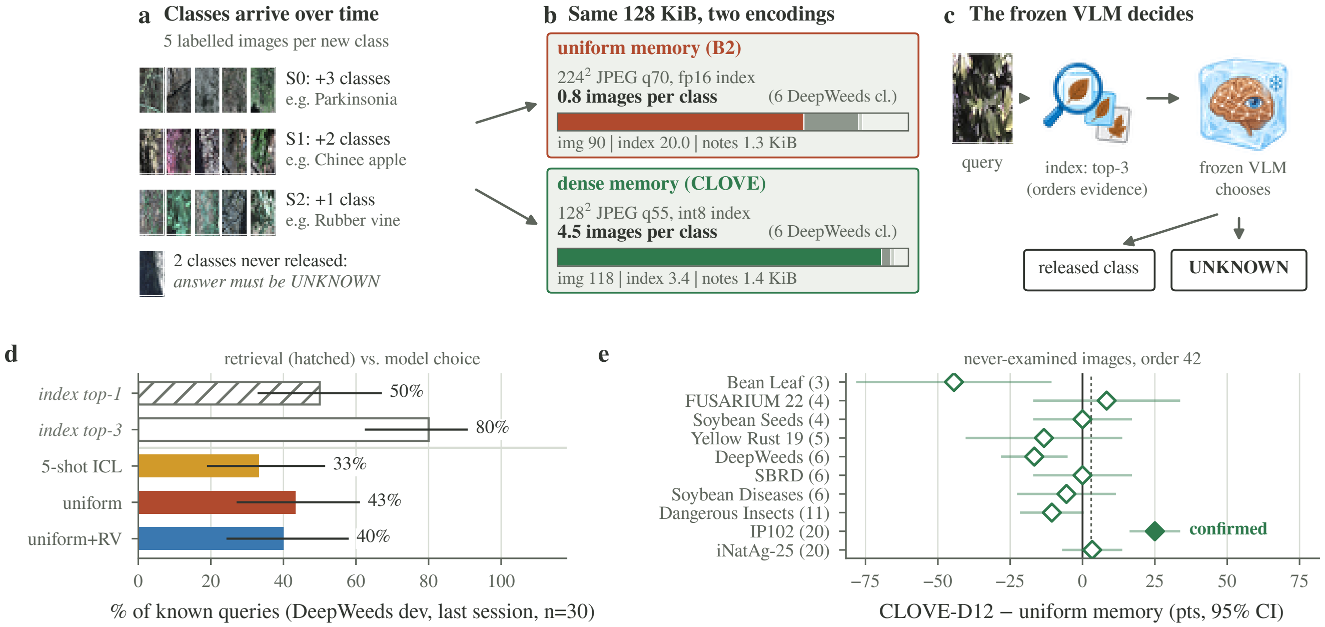


Figure 1. CLOVE at a glance. (a) Agricultural classes are released over sessions with five labelled images each (real DeepWeeds supports, class order 42); held-out classes are never released and must be answered UNKNOWN. (b) Both memories have the same 128 KiB budget. Bars show the measured bytes at the last session (images, index, notes); the denser encoding retains more references per class, and the realized count depends on the class inventory and image content (Fig. 4a). (c) The memory index only ranks and orders evidence; the frozen VLM chooses a released class or UNKNOWN. (d) Retrieval versus decision on DeepWeeds development data: the true class is in the index top three far more often than the model's forced choice is correct (Wilson 95% intervals). (e) Pre-registered confirmation on never-examined images (order 42): paired effect of CLOVE-D12 against the uniform memory with unadjusted 95% CI; filled marker: confirmed under the rule (≥ 3 points, CI above zero, $p_{\text{Holm}} < 0.05$). Released classes in brackets. Photographs are real data; icons are illustrations.

Abstract

CLOVE (Continual Learning for Open-world Visual Evaluation in agriculture) asks whether a small, locally runnable vision-language model can itself be the classifier when agricultural categories are released over time with five labelled images each, while images of never-taught categories keep arriving. Frozen Qwen2.5-VL-3B-Instruct [3] (arXiv 2025) chooses a released class or UNKNOWN; nothing is trained and only a 128 KiB memory changes. On identical supports, class orders and queries we find three things. (i) Generative collapse: as classes accumulate the model's decision degrades although retrieval covers the answer—on DeepWeeds development data the true class is in the index top three for 80% of images, but the forced choice is right for 43%. (ii) Agentic orchestration does not repair it: under pre-registered confirmation on never-examined image groups, exhaustive class verification is not confirmed and retrieve-then-verify is not con-

firmed. (iii) Memory optimisation helps where the byte budget is the bottleneck: a dense memory that stores several times more references per class, combined with retrieve-then-verify (CLOVE-D12), meets the pre-registered rule on 2 of 11 data sets (IP102 (25 cl.) (+25 points vs B2); PlantDoc (+27 points vs B12)), is lower on 5 and otherwise indistinguishable. Frozen-feature readouts such as FeCAM [7] (NeurIPS 2023) and RanPAC [13] (NeurIPS 2023) remain more accurate in closed-set accuracy than any generative arm. All arms, including every failed one, are reported.

1. Introduction

A recognition system for crops, weeds and pests is updated as new categories become relevant; each update brings only a few labelled images, and images of categories that were never taught keep arriving. A frozen vision-language model (VLM) with an external memory is an attractive design for this setting: nothing is trained, the memory is inspectable, and the same model also answers free-form questions. Benchmarks such as AgEval [1] (WACV 2025) evaluate VLMs on fixed agricultural label sets, and agentic systems such as SAGE [2] (arXiv 2026) ground diagnoses in retrieved knowledge; neither releases classes over time under a memory budget, and neither lets a small local model act as the continual classifier.

We study exactly this setting and follow one line of evidence (Fig. 1). First, the model's own multiple-choice decision collapses as classes accumulate, although the memory index still ranks the true class near the top; predictions concentrate on a few attractive names and change with option order (Sec. 5.1). Second, the usual agentic remedies—SAGE-style inspection, retrieve-then-verify and exhaustive per-class verification—do not survive pre-registered confirmation (Sec. 5.2). Third, we optimise what the memory stores within the same byte budget, select the memory–orchestration combination on development data only, and confirm it on never-examined images of every eligible data set (Sec. 5.3). Fourth, we follow the result data set by data set (Sec. 5.4).

Key finding

The bottleneck of a frozen generative classifier is its decision among names, not visual coverage. Orchestration does not move it; storing more, smaller references per class does—but only where many classes compete for the budget.

Contributions. (1) A training-free, byte-bounded, open-world class-incremental protocol in which the VLM makes every decision, on 11 agricultural data sets, with a coverage audit of 21 candidates. (2) A diagnosis of generative collapse (coverage versus decision, attractors, order sensitivity). (3) Pre-registered negative results for agentic orchestration and verification. (4) A byte-optimised memory and a data-set-level account of when it helps. No component is claimed as new by itself.

2. Related Work

Agricultural VLM evaluation. AgEval [1] (WACV 2025) and AgroBench [18] (ICCV 2025) evaluate VLMs on static agricultural tasks; SAGE [2] (arXiv 2026) and AgriRAG [12] (CVPRW 2026) retrieve knowledge or images for a VLM. None releases categories over time or bounds memory.

Retrieval and in-context recognition. RAR [11] (arXiv 2024), Visual RAG [5] (arXiv 2025) and in-context classification with multimodal models [6, 9] (arXiv 2026, 2024) are the closest generic techniques; open-weight models gain little from many in-context examples [9].

Continual and open-world learning. FSCIL [19] (CVPR 2020), SLDA [8] (CVPRW 2020), FeCAM [7] (NeurIPS 2023), RanPAC [13] (NeurIPS 2023), TEEN [20] (NeurIPS 2023) and L2P [22] (CVPR 2022) learn readouts or prompts; InCA [15] (arXiv 2024) pairs a frozen language model with an external learner; open-world recognition [4] (CVPR 2015) and open-world continual learning [10] (arXiv 2023) add rejection. Agent memories with consolidation [16, 17] (UIST 2023, arXiv 2023) and description-based classification [14] (ICLR 2023) motivate our memory notes.

3. Preliminaries

Setting. After session t the released inventory is $C_t = C_{t-1} \cup \Delta C_t$; each new class comes with $K = 5$ labelled supports S_t. The model is frozen, $\theta_t = \theta_0$, and only the memory evolves:

$$M_t = \mathcal{C}_B\big(M_{t-1} \cup E(S_t)\big), \qquad \text{bytes}(M_t) \le B = 131{,}072, \tag{1}$$

where E writes a note with the frozen model from a class's own supports and $\mathcal{C}_B$ evicts stored references until the budget holds. A query is answered $\hat{y} \in C_t \cup \{\text{UNKNOWN}\}$ by the model. Future and held-out classes are never named, shown or stored.

Decision rule. Every arm ends with one constrained call listing the released classes and UNKNOWN as letters; the label is the letter with the highest next-token probability. Known-ness is $1-p(\text{UNKNOWN})$. Open-world balanced accuracy (BA, primary) counts UNKNOWN on a known image as an error; forced-choice BA removes UNKNOWN; calibrated BA answers UNKNOWN only below the known-ness that keeps 95% of known development images.

Data and cohorts. Locked CLOVE-Bench manifests (group-aware, de-duplicated by SHA-256 and perceptual hash) fix supports and five class orders. Development queries come from the locked development split; confirmatory (fresh) queries come from source groups used by no earlier experiment, at most one per group. Table 1 lists every candidate data set and its status.

4. CLOVE: Memory and Orchestration

Uniform memory (B2). A class record holds a note ($\leq$48 tokens), 224^2 JPEG thumbnails (quality 70) of retained supports and an fp16 index vector each; eviction removes thumbnails from the class with most thumbnails. A 224^2 textured field image costs about 17 KB, so under 128 KiB the memory keeps about one image per class or fewer once ten or more classes are released (Fig. 4a).

Dense memory (D). Same budget and eviction rule, but 128^2 thumbnails (quality 55, $\approx$5 KB) and an int8 code of a fixed random projection of the frozen vision feature to 128 dimensions, regenerated from a stored seed (130 bytes per thumbnail instead of 4 KB). The index only orders evidence; the model makes every decision.

Orchestration. B0 zero-shot; B1 five-shot in-context learning (two raw supports per class); B2 one prompt with the memory; B3 fixed SAGE-style orchestration (text shortlist, then inspection of three classes); B4 autonomous orchestration (the model chooses which classes to inspect); B12 retrieve-then-verify (the index shortlists three classes, the model chooses among them or UNKNOWN); C1 exhaustive independent verification of every class followed by a final call. CLOVE-D12 is retrieve-then-verify on the dense memory; it was selected among D2, D3, D4 and D12 by a pre-stated rule on development data only (DeepWeeds and Mango).

5. Experiments

Setup. Qwen2.5-VL-3B-Instruct [3] (arXiv 2025) (revision 66285546, fp16) on two 12 GB GPUs; at most 12 thumbnails per prompt. Statistics: class-stratified paired bootstrap over source groups (orders crossed when several are pooled), exact McNemar, Holm within each pre-registered family. A contrast is confirmed with $\Delta \geq 3$ points, 95% CI above zero and $p_{\text{Holm}} < 0.05$.

Frozen-feature reference. Table 2 reports continual readouts on the same Qwen vision features and manifests (closed-set accuracy on the locked test split, final session). They decide labels from features and are therefore not CLOVE arms, but they show what the frozen representation supports.

5.1. Generative collapse

At the last DeepWeeds development session the true class is in the index top three for 80% of images, but the model's forced choice is correct for 43% with the memory, 33% with five-shot in-context learning and 40% when only the three retrieved classes are offered (Fig. 1d, Fig. 2). The most-predicted label receives 33% of known images where 17% is expected, and reversing the option order preserves only 50% of answers. A blank query image yields UNKNOWN for every known image and permuting class names against the evidence lowers accuracy, so the evidence is used.

Observation

Coverage is not the problem: the answer is retrieved, but the frozen model does not pick it.

5.2. Agentic orchestration under pre-registered confirmation

On never-examined groups of DeepWeeds and Dangerous Insects, registered before any fresh image was evaluated, exhaustive verification (C1) is not confirmed, retrieve-then-verify (B12) is not confirmed and five-shot in-context learning (B1) is not confirmed relative to the uniform memory (Fig. 3, upper block).

5.3. Memory optimisation

On development data the dense memory lifts DeepWeeds forced choice when combined with retrieve-then-verify, and the pre-stated rule selected D12. On never-examined groups (Table 3) the pre-registered rule is met on 1 of 10 data sets (IP102 (25 cl.)). Calibrated rejection of held-out classes is reported alongside, because the dense arm answers UNKNOWN more often in open-world mode.

5.4. Data set by data set

Table 4 and Fig. 5 place all data sets on one axis: how many classes compete for the 128 KiB budget. Where

Table 1. Data-set coverage: all twelve official AgEval subsets, further identification sets of the AgEval loader, and external benchmarks. “evaluated”: part of the pre-registered confirmation.

Data set	Classes	Source	Status	Note
AgEval				
Soybean Seeds	5	Kaggle (AgEval loader)	evaluated	
Mango Leaf Disease	8	Kaggle MangoLeafBD (AgEval loader)	development only	almost no source groups outside earlier partitions
DeepWeeds	8	Kaggle DeepWeeds (AgEval loader)	evaluated	Negative class excluded
Bean Leaf Lesions	3	Kaggle (AgEval loader)	evaluated	incremental only (no held-out class)
Dangerous Insects	15	Kaggle (AgEval loader)	evaluated	
Soybean Diseases (PNAS)	7	Zenodo 12747481 (AgEval loader)	evaluated	folders 1/7 excluded (label-map conflict)
Durum Wheat	3	Kaggle (AgEval loader)	excluded	each class is frames of one video: 5 independent supports impossible
Yellow Rust 19	6	AgEval loader	evaluated	severity grades as classes (ordinal caveat); RAW photos, augmented copies avoided
FUSARIUM 22	5	AgEval loader	evaluated	wilt grades as classes (ordinal caveat)
IDC (iron deficiency chlorosis)	4	AgEval loader	development only	severity ratings; ratings 3-4 have no never-examined groups; rating 5 insufficient
PlantDoc (segmentation)	–	AgEval loader	excluded	segmentation/quantification task: no class labels
InsectCount	–	AgEval loader	excluded	counting task: no class labels
AgEval loader				
Rice Leaf Disease Severity (SBRD)	9	Kaggle (AgEval loader)	evaluated	
IP102 (25 classes)	25	Kaggle IP102 (AgEval loader)	evaluated	25 hash-selected classes
external				
iNatAg-25	25	HF Project-AgML/iNatAg-mini	evaluated	
iNatAg-100	100	HF Project-AgML/iNatAg-mini	running	pre-registered >=25-class extension
PlantDoc (classification)	27	GitHub pratikkayal/PlantDoc-Dataset (CC BY 4.0)	evaluated	web images; spider-mite class excluded (2 images)
PlantVillage	38	HF mohanty/PlantVillage (CC BY-SA 3.0)	running	grouped by leaf-map; 26% ungrouped
IP102 (96 classes)	96	Kaggle IP102	running	6 classes with <60 images excluded; overlap with IP102-25 cohorts excluded
PlantWild	89	HF uqtwei2/PlantWild (CC BY-NC-ND 4.0)	development only	40/class sample used every source group; PlantWild-v2 not used (no split metadata)
iNatAg-500	500	HF Project-AgML/iNatAg-mini	not run	46 GB stream exceeds the run budget

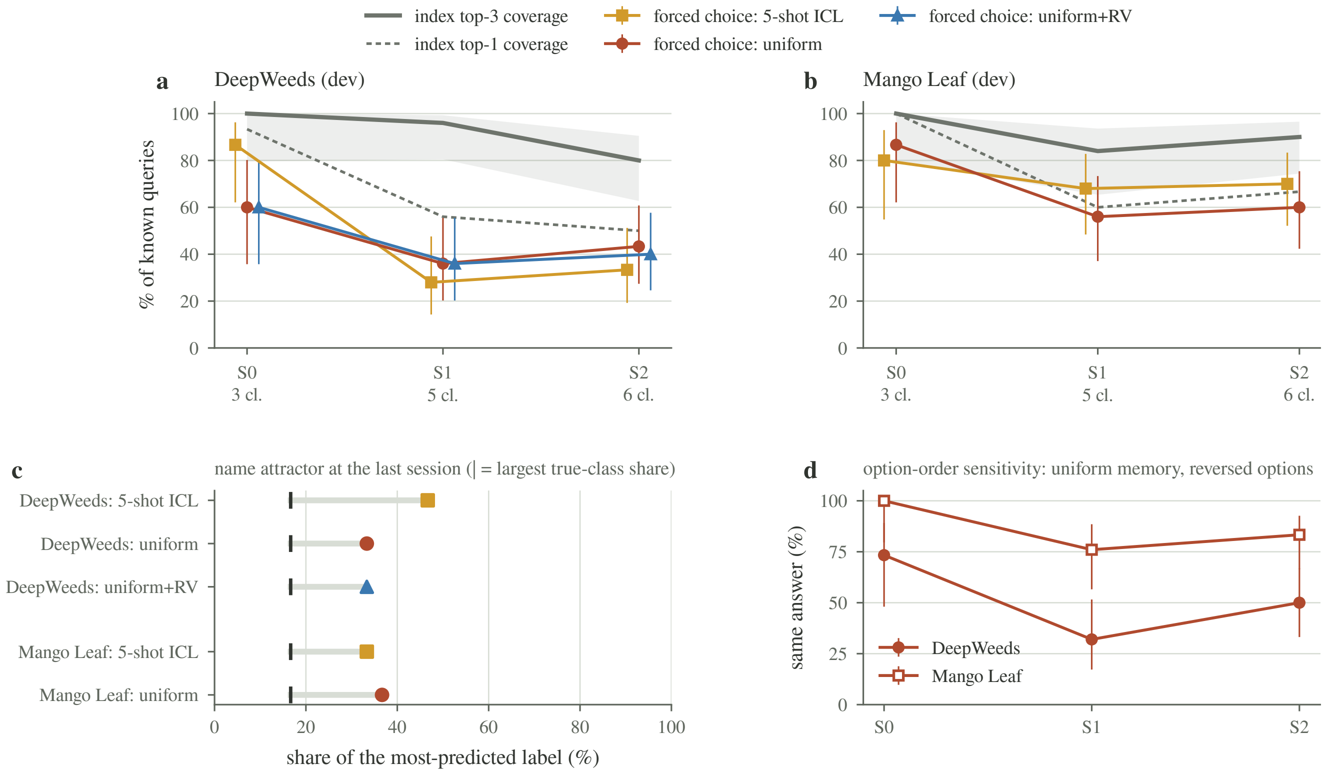
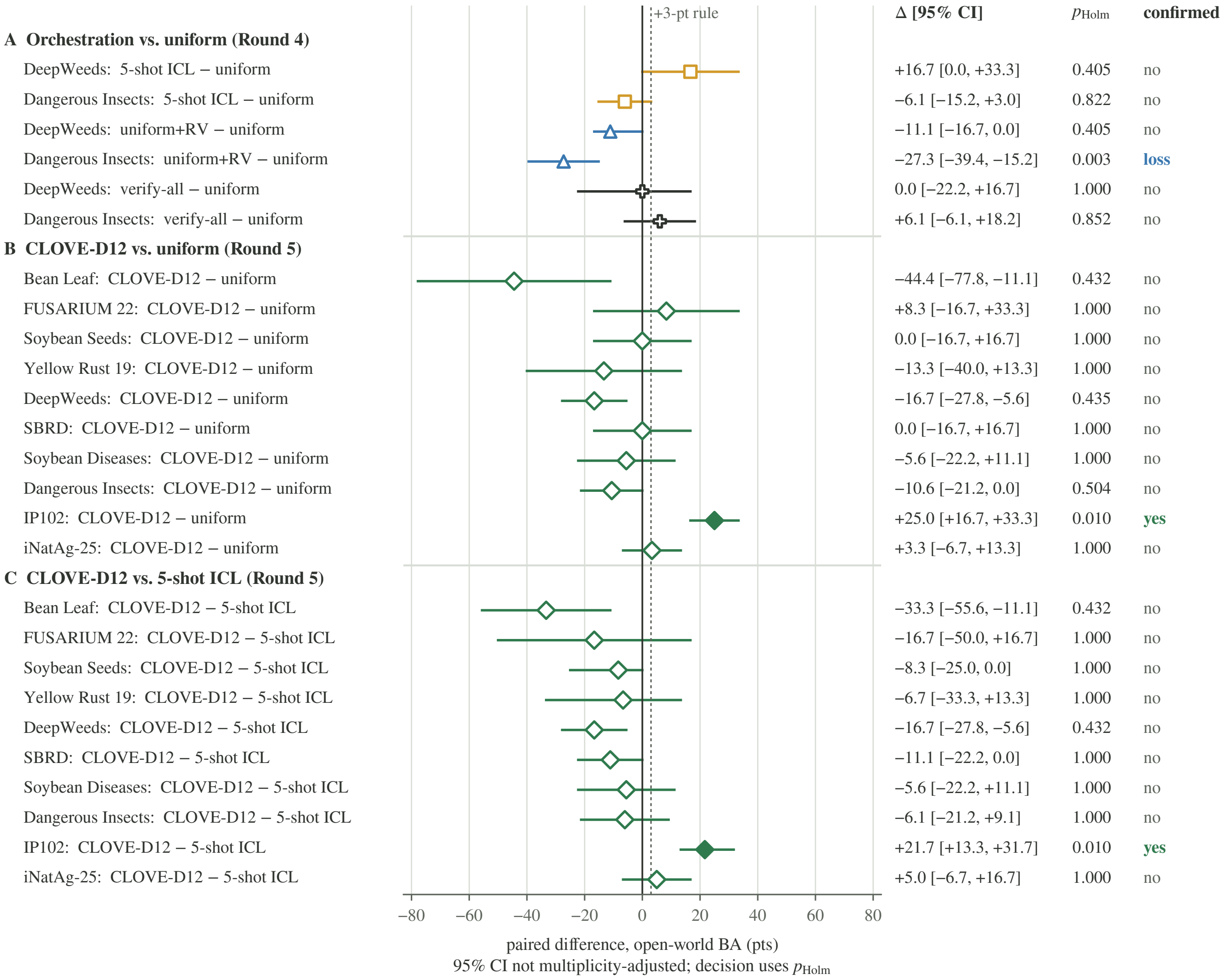


Figure 2. Does the model fail because evidence is missing, or because it does not use it? Development cohort, class order 42. (a,b) Grey: retrieval quantities (share of known queries whose class is in the memory index’s top-1 / top-3; band: Wilson 95% interval for top-3). Colour: the model’s forced-choice accuracy on the same queries (Wilson 95%). (c) Share of known queries assigned to the most-predicted label at the last session, against the largest true-class share. (d) Agreement of the uniform-memory answer with the answer under reversed option order.

few classes are released the uniform memory already holds two to three images per class and the dense memory adds little or hurts; where twenty or more classes are released the uniform memory falls below one image per class.

Key finding

Memory optimisation pays off only where the budget is the bottleneck (IP102 (25 cl.) (+25 points vs B2); PlantDoc (+27 points vs B12)); on few-class data sets the uniform memory already fits and the dense memory adds noise.



	Δ [95% CI]	p_{Holm}	confirmed
A Orchestration vs. uniform (Round 4)			
DeepWeeds: 5-shot ICL – uniform	+16.7 [0.0, +33.3]	0.405	no
Dangerous Insects: 5-shot ICL – uniform	−6.1 [−15.2, +3.0]	0.822	no
DeepWeeds: uniform+RV – uniform	−11.1 [−16.7, 0.0]	0.405	no
Dangerous Insects: uniform+RV – uniform	−27.3 [−39.4, −15.2]	0.003	**loss**
DeepWeeds: verify-all – uniform	0.0 [−22.2, +16.7]	1.000	no
Dangerous Insects: verify-all – uniform	+6.1 [−6.1, +18.2]	0.852	no
B CLOVE-D12 vs. uniform (Round 5)			
Bean Leaf: CLOVE-D12 – uniform	−44.4 [−77.8, −11.1]	0.432	no
FUSARIUM 22: CLOVE-D12 – uniform	+8.3 [−16.7, +33.3]	1.000	no
Soybean Seeds: CLOVE-D12 – uniform	0.0 [−16.7, +16.7]	1.000	no
Yellow Rust 19: CLOVE-D12 – uniform	−13.3 [−40.0, +13.3]	1.000	no
DeepWeeds: CLOVE-D12 – uniform	−16.7 [−27.8, −5.6]	0.435	no
SBRD: CLOVE-D12 – uniform	0.0 [−16.7, +16.7]	1.000	no
Soybean Diseases: CLOVE-D12 – uniform	−5.6 [−22.2, +11.1]	1.000	no
Dangerous Insects: CLOVE-D12 – uniform	−10.6 [−21.2, 0.0]	0.504	no
IP102: CLOVE-D12 – uniform	+25.0 [+16.7, +33.3]	0.010	**yes**
iNatAg-25: CLOVE-D12 – uniform	+3.3 [−6.7, +13.3]	1.000	no
C CLOVE-D12 vs. 5-shot ICL (Round 5)			
Bean Leaf: CLOVE-D12 – 5-shot ICL	−33.3 [−55.6, −11.1]	0.432	no
FUSARIUM 22: CLOVE-D12 – 5-shot ICL	−16.7 [−50.0, +16.7]	1.000	no
Soybean Seeds: CLOVE-D12 – 5-shot ICL	−8.3 [−25.0, 0.0]	1.000	no
Yellow Rust 19: CLOVE-D12 – 5-shot ICL	−6.7 [−33.3, +13.3]	1.000	no
DeepWeeds: CLOVE-D12 – 5-shot ICL	−16.7 [−27.8, −5.6]	0.432	no
SBRD: CLOVE-D12 – 5-shot ICL	−11.1 [−22.2, 0.0]	1.000	no
Soybean Diseases: CLOVE-D12 – 5-shot ICL	−5.6 [−22.2, +11.1]	1.000	no
Dangerous Insects: CLOVE-D12 – 5-shot ICL	−6.1 [−21.2, +9.1]	1.000	no
IP102: CLOVE-D12 – 5-shot ICL	+21.7 [+13.3, +31.7]	0.010	**yes**
iNatAg-25: CLOVE-D12 – 5-shot ICL	+5.0 [−6.7, +16.7]	1.000	no

Figure 3. Every confirmatory contrast, by inferential family. Paired difference in final-session open-world balanced accuracy on never-examined image groups with 95% bootstrap CIs that are not multiplicity-adjusted; the decision uses Holm-adjusted p within each pre-registered family (A: the six Round-4 orchestration tests; B and C together: one family of all Round-5 order-42 tests). "yes": ≥ 3 points, CI above zero and $p_{\text{Holm}} < 0.05$; "loss": CI below zero and $p_{\text{Holm}} < 0.05$. An unadjusted CI excluding zero does not by itself imply significance after correction: five Round-5 contrasts (Bean Leaf, DeepWeeds, Dangerous Insects) have CIs below zero but $p_{\text{Holm}} \geq 0.336$, and exact McNemar tests on their 9–30 paired units are also non-significant.

6. Limitations and Conclusion

Limitations. Query sets per data set are small and the primary test uses one class order; further orders are pooled where complete. One 3B model in fp16 was used. Mango Leaf serves for development only. Pretraining exposure to these public data sets cannot be excluded. The many-class hypothesis was formed after the IP102 result; its extension to iNatAg-100 is registered separately. Monolithic prompts are limited to 25 classes by the single-token option alphabet.

Not completed. Memory-budget sweeps (32 KB–1 MB), shot-count sweeps ($K \in \{1, 2, 4, 8, 16\}$), domain-shift tracks, the remaining AgEval severity subsets and external data sets (PlantDoc, PlantVillage) were not completed within the compute budget; no result is reported for them.

Conclusion. A frozen 3B VLM with a 128 KiB external memory learns new agricultural categories from five images while few categories are known, but its own decision collapses as categories accumulate even when re-

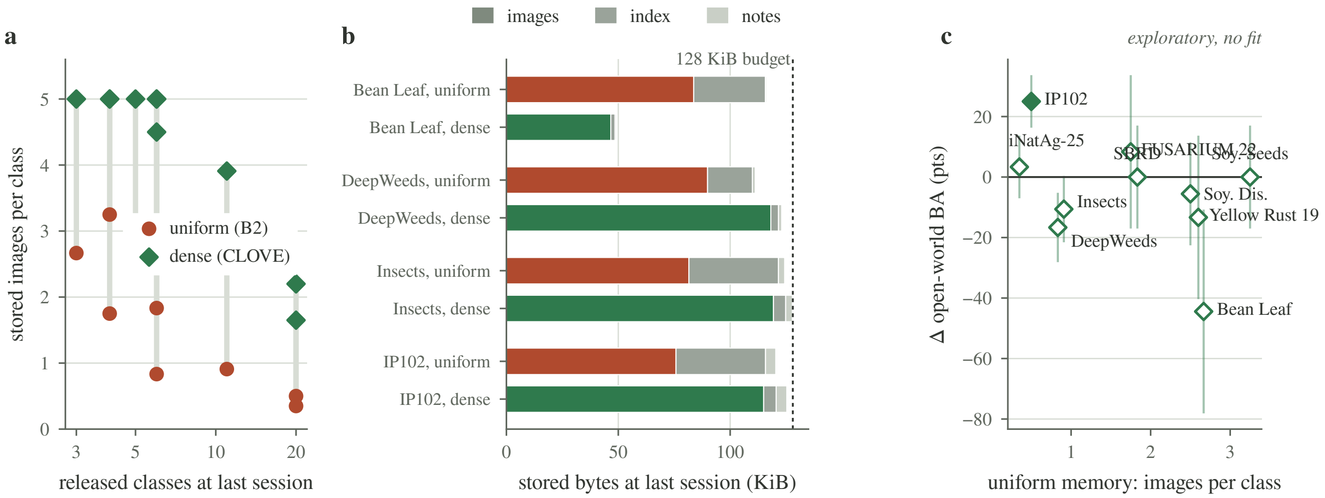


Figure 4. Fixed-budget memory analysis (128 KiB; no budget sweep is reported). (a) Measured stored images per class at the last session (order 42) for each data set, against the number of released classes. (b) Measured byte composition; on Bean Leaf the dense memory holds every support well below the budget, so its loss there is not a budget effect. (c) Exploratory: CLOVE-D12 effect against how starved the uniform memory is (no fit; too few independent data sets for a trend claim).

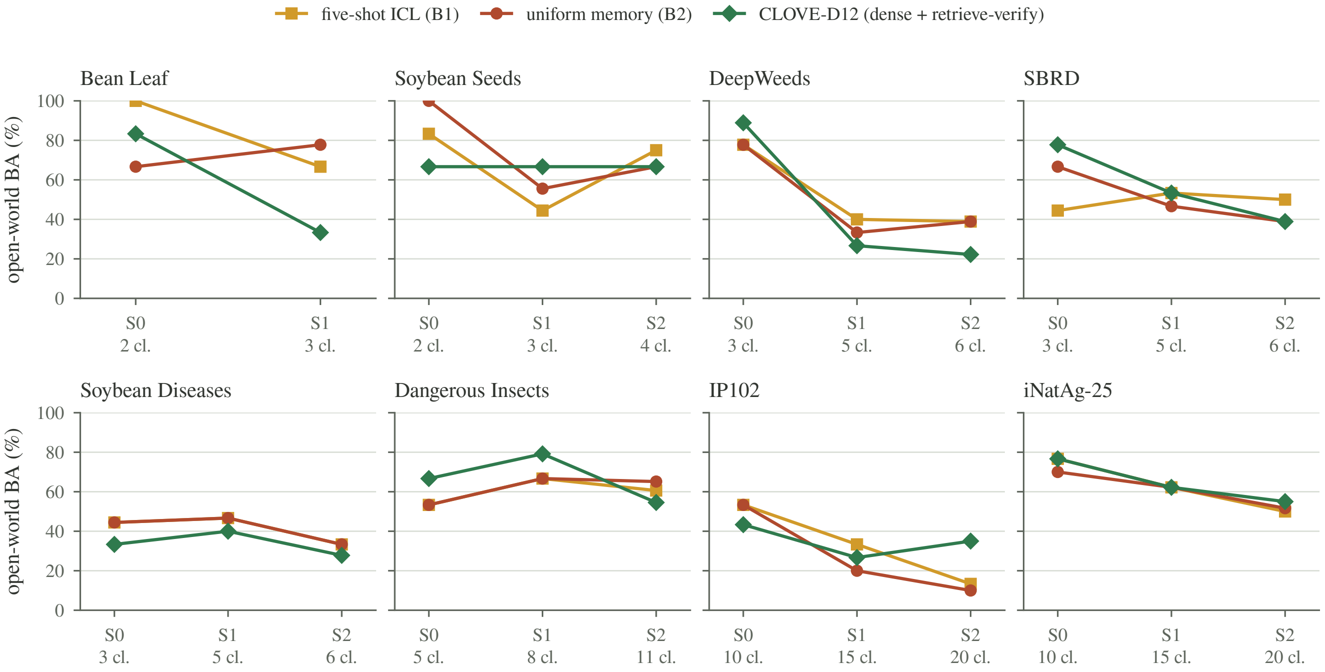


Figure 5. Continual learning curves. Open-world balanced accuracy at every session on never-examined image groups, one class order (42); the tick labels give the number of released classes. Bean Leaf has two sessions. Colours and markers are fixed per method in every figure.

trieval covers the answer. Agentic inspection and verification do not repair this. Optimising what the memory stores helps only where the byte budget binds, and frozen-feature readouts remain stronger in closed-set accuracy. Closing the gap between what the memory contains and what the model decides is the open problem.

Table 2. Frozen-feature continual readouts on the same Qwen vision features (closed-set accuracy, %, final session, locked test split; best in bold). NCM: nearest class mean [21] (arXiv 2019); SLDA [8] (CVPRW 2020); FeCAM [7], RanPAC [13], TEEN [20] (all NeurIPS 2023); Linear: full-memory linear probe.

Data set	NCM	1-NN	SLDA	SLDA-r8	FeCAM	RanPAC	TEEN	Linear
Bean Leaf	76	69	82	76	73	77	68	75
DeepWeeds	72	68	74	72	74	76	68	74
Dangerous Insects	70	65	72	71	64	73	66	72
Mango Leaf	82	84	93	92	89	90	77	82
SBRD	68	72	74	73	72	75	65	71
Soybean Seeds	66	52	66	70	59	68	61	66
iNatAg-25	78	77	83	83	77	83	76	82

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Table 3. CLOVE-D12 against the uniform memory (B2) and five-shot ICL (B1) on never-examined image groups (final session, mean over completed class orders). BA open / forced / calibrated, old / new classes, held-out rejection (raw and calibrated), known-ness AUROC, model calls per query.

Arm	BA	BA$_f$	BA$_{\text{cal}}$	Old	New	Rej.	Rej.$_{\text{cal}}$	AUROC	Calls
DeepWeeds									
B1 five-shot ICL (raw supports)	39	50	41	38	44	39	11	0.60	1.0
B2 uniform memory	41	44	38	40	44	39	28	0.64	1.0
D12 dense + retrieve-then-verify	26	48	36	27	22	72	56	0.71	1.0
Dangerous Insects									
B1 five-shot ICL (raw supports)	61	67	61	67	44	67	58	0.83	1.0
B2 uniform memory	65	68	65	67	61	75	75	0.87	1.0
D12 dense + retrieve-then-verify	55	58	58	58	44	92	92	0.83	1.0
SBRD									
B1 five-shot ICL (raw supports)	58	64	59	50	100	17	0	0.58	1.0
B2 uniform memory	53	53	50	53	50	0	0	0.52	1.0
D12 dense + retrieve-then-verify	44	58	52	40	67	67	39	0.77	1.0
Soybean Seeds									
B1 five-shot ICL (raw supports)	71	79	80	67	83	50	0	0.78	1.0
B2 uniform memory	71	71	77	61	100	0	0	0.62	1.0
D12 dense + retrieve-then-verify	58	67	60	50	83	33	33	0.78	1.0
Bean Leaf									
B1 five-shot ICL (raw supports)	56	56	56	58	50	—	—	—	1.0
B2 uniform memory	72	72	72	83	50	—	—	—	1.0
D12 dense + retrieve-then-verify	50	50	50	75	0	—	—	—	1.0
iNatAg-25									
B1 five-shot ICL (raw supports)	53	56	56	54	49	36	4	0.67	1.0
B2 uniform memory	50	52	53	56	31	44	4	0.70	1.0
D12 dense + retrieve-then-verify	56	62	64	56	53	58	7	0.67	1.0
IP102 (25 cl.)									
B1 five-shot ICL (raw supports)	25	26	26	24	29	49	18	0.66	1.0
B2 uniform memory	14	16	15	16	11	67	16	0.63	1.0
D12 dense + retrieve-then-verify	31	48	43	27	44	69	36	0.66	1.0
Soybean Diseases (7 cl.)									
B1 five-shot ICL (raw supports)	36	39	38	33	50	0	0	0.74	1.0
B2 uniform memory	33	33	33	30	50	0	0	0.89	1.0
D12 dense + retrieve-then-verify	39	50	43	30	83	17	0	0.54	1.0
PlantDoc									
D12 dense + retrieve-then-verify	47	53	52	44	58	20	7	0.56	1.0
B12 uniform + retrieve-then-verify	20	20	20	24	0	27	0	0.48	1.0
Yellow Rust 19									
B1 five-shot ICL (raw supports)	40	40	33	50	0	0	0	0.84	1.0
B2 uniform memory	47	47	40	58	0	0	0	0.82	1.0
D12 dense + retrieve-then-verify	33	33	27	33	33	100	33	1.00	1.0
FUSARIUM 22									
B1 five-shot ICL (raw supports)	50	50	42	44	67	0	0	0.58	1.0
B2 uniform memory	25	25	25	11	67	0	0	0.36	1.0
D12 dense + retrieve-then-verify	33	42	33	44	0	67	0	0.44	1.0

Table 4. Storyline across data sets (never-examined groups, final session). $|C_T|$ released classes; Refs: stored images per class in the uniform (U) and dense (D) memory; BA: open-world balanced accuracy (%), mean over completed orders; Δ: paired CLOVE-D12 effect on order 42 with unadjusted 95% CI; verdict uses p_{Holm} and the pre-registered rule. Total: classes in the locked subset; Held: never-released classes; —: not applicable (arm not run).

Data set	Total	Held	$\lvert C_T\rvert$	Refs U	Refs D	B1	B2	B12	D12	D12$_f$	Δ (pts)	p_{Holm}	Verdict
Bean Leaf	3	0	3	2.7	5.0	56	72	—	50	50	−44 [−78, −11] (vs B2)	0.432	lower (n.s. after Holm)
Soybean Seeds	5	1	4	3.2	5.0	71	71	—	58	67	+0 [−17, +17] (vs B2)	1.000	no detectable difference
FUSARIUM 22	5	1	4	1.8	5.0	50	25	—	33	42	+8 [−17, +33] (vs B2)	1.000	no detectable difference
Yellow Rust 19	6	1	5	2.6	5.0	40	47	—	33	33	−13 [−40, +13] (vs B2)	1.000	no detectable difference
SBRD	9	3	6	1.8	5.0	58	53	—	44	58	+0 [−17, +17] (vs B2)	1.000	no detectable difference
DeepWeeds	8	2	6	0.8	4.5	39	41	—	26	48	−17 [−28, −6] (vs B2)	0.435	lower (n.s. after Holm)
Soybean Diseases (7 cl.)	7	1	6	2.5	5.0	36	33	—	39	50	−6 [−22, +11] (vs B2)	1.000	no detectable difference
Dangerous Insects	15	4	11	0.9	3.9	61	65	—	55	58	−11 [−21, −0] (vs B2)	0.504	lower (n.s. after Holm)
IP102 (25 cl.)	25	5	20	0.5	2.2	25	14	—	31	48	+25 [+17, +33] (vs B2)	0.010	confirmed gain
iNatAg-25	25	5	20	0.3	1.6	53	50	—	56	62	+3 [−7, +13] (vs B2)	1.000	no detectable difference
PlantDoc	27	5	22	0.4	1.9	—	—	20	47	53	+27 [+18, +36] (vs B12)	0.002	confirmed gain

A. Dataset-by-Dataset Account

Bean Leaf (3 released classes at the last session; the uniform memory keeps 2.7 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 56% with five-shot in-context learning, 72% with the uniform memory and 50% with D12 (forced choice 72% vs 50%). D12−B2 on order 42: −44 [−78, −11] points, $p_{\text{Holm}} = 0.432$: lower (n.s. after Holm).

Soybean Seeds (4 released classes at the last session; the uniform memory keeps 3.2 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 71% with five-shot in-context learning, 71% with the uniform memory and 58% with D12 (forced choice 71% vs 67%). D12−B2 on order 42: +0 [−17, +17] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 33% (D12).

FUSARIUM 22 (4 released classes at the last session; the uniform memory keeps 1.8 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 50% with five-shot in-context learning, 25% with the uniform memory and 33% with D12 (forced choice 25% vs 42%). D12−B2 on order 42: +8 [−17, +33] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 0% (D12).

Yellow Rust 19 (5 released classes at the last session; the uniform memory keeps 2.6 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 40% with five-shot in-context learning, 47% with the uniform memory and 33% with D12 (forced choice 47% vs 33%). D12−B2 on order 42: −13 [−40, +13] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 33% (D12).

SBRD (6 released classes at the last session; the uniform memory keeps 1.8 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 58% with five-shot in-context learning, 53% with the uniform memory and 44% with D12 (forced choice 53% vs 58%). D12−B2 on order 42: +0 [−17, +17] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 39% (D12).

DeepWeeds (6 released classes at the last session; the uniform memory keeps 0.8 and the dense memory 4.5 thumbnails per class). Open-world balanced accuracy is 39% with five-shot in-context learning, 41% with the uniform memory and 26% with D12 (forced choice 44% vs 48%). D12−B2 on order 42: −17 [−28, −6] points, $p_{\text{Holm}} = 0.435$: lower (n.s. after Holm). Calibrated held-out rejection: 28% (B2) vs 56% (D12).

Soybean Diseases (7 cl.) (6 released classes at the last session; the uniform memory keeps 2.5 and the dense memory 5.0 thumbnails per class). Open-world balanced accuracy is 36% with five-shot in-context learning, 33% with the uniform memory and 39% with D12 (forced choice 33% vs 50%). D12−B2 on order 42: −6 [−22, +11] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 0% (B2) vs 0% (D12).

Dangerous Insects (11 released classes at the last session; the uniform memory keeps 0.9 and the dense memory 3.9 thumbnails per class). Open-world balanced accuracy is 61% with five-shot in-context learning, 65% with the uniform memory and 55% with D12 (forced choice 68% vs 58%). D12−B2 on order 42: −11 [−21, −0] points, $p_{\text{Holm}} = 0.504$: lower (n.s. after Holm). Calibrated held-out rejection: 75% (B2) vs 92% (D12).

IP102 (25 cl.) (20 released classes at the last session; the uniform memory keeps 0.5 and the dense memory 2.2 thumbnails per class). Open-world balanced accuracy is 25% with five-shot in-context learning, 14% with the uniform memory and 31% with D12 (forced choice 16% vs 48%). D12−B2 on order 42: +25 [+17, +33] points, $p_{\text{Holm}} = 0.010$: confirmed gain. Calibrated held-out rejection: 16% (B2) vs 36% (D12).

iNatAg-25 (20 released classes at the last session; the uniform memory keeps 0.3 and the dense memory 1.6 thumbnails per class). Open-world balanced accuracy is 53% with five-shot in-context learning, 50% with the uniform memory and 56% with D12 (forced choice 52% vs 62%). D12−B2 on order 42: +3 [−7, +13] points, $p_{\text{Holm}} = 1.000$: no detectable difference. Calibrated held-out rejection: 4% (B2) vs 7% (D12).

PlantDoc (22 released classes at the last session; the uniform memory keeps 0.4 and the dense memory 1.9 thumbnails per class). Open-world balanced accuracy is 20% with retrieve-then-verify on the uniform memory and 47% on the dense memory. D12−B12 on order 42: +27 [+18, +36] points, $p_{\text{Holm}} = 0.002$: confirmed gain.

B. Development Results and Gap Analysis

C. Round-4 Confirmation Details

The confirmatory run was registered before any fresh image was evaluated (rounds/R4/preregistration.md): DeepWeeds and Dangerous Insects, class order 42, three known and three held-out queries per class and session from source groups never examined before, arms B0, B1, B2, B12 and C1, primary endpoint final-session open-world balanced accuracy, and the family {C1−B2, B12−B2, B1−B2} on both data sets with Holm correction. A contrast is confirmed only with $\Delta \geq 3$ points, a 95% interval above zero, $p_{\text{Holm}} < 0.05$ on both data sets, and no more than 10 points lower held-out rejection. Under this rule independent verification (C1) is not confirmed, retrieve-then-verify (B12) is not confirmed and five-shot in-context learning (B1) is not confirmed relative to the uniform memory (B2). Tables 7 and 8 and Figure 6 give all values.

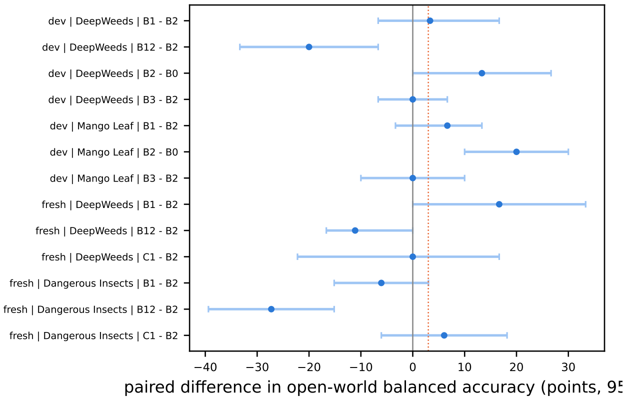

Figure 6. Paired differences in open-world balanced accuracy with 95% intervals (development and confirmatory). Dotted line: the pre-registered +3-point threshold.

D. Round-5 Details

E. Step 3: Memory Optimisation (Round 5)

The development diagnostics show that the 128 KiB uniform memory keeps 0–1 reference image per DeepWeeds class, because a 224^2 JPEG of a textured weed image is about 17 KB and its fp16 index another 4 KB. The dense memory keeps the same budget and eviction rule but stores 128^2 thumbnails (quality 55, about 5 KB) and replaces the fp16 index by an int8 code of a fixed random projection to 128 dimensions that is regenerated from a stored seed (130 bytes per thumbnail). The index orders and shortlists evidence only; the model chooses every label. We combine it with the three orchestration designs: D2 (monolithic), D3 (fixed SAGE-style), D4 (autonomous) and D12 (retrieve-then-verify), with B2 (uniform memory) and B1 (five-shot ICL on raw supports) in the same runs. A secondary, pre-declared calibrated read-out takes the model's forced choice and answers UNKNOWN only when its known-ness falls below the threshold that keeps 95% of known development images of the same arm and data set.

Confirmation. The selected arm D12 was compared with B2 and B1 on never-examined image groups of DeepWeeds, Dangerous Insects, SBRD, Soybean Seeds, iNatAg-25 and Bean Leaf, plus the two AgEval data sets added afterwards, IP102 (25 classes) and Soybean

Table 5. Development cohort, class order 42, final session (columns as in Table 3; index-only row is not a model decision).

Arm	BA	BA$_f$	Old	New	Rej.	FR	AUROC	Calls
Mango Leaf, final session 2, 30 known / 9 held-out queries								
B0 zero-shot	40	40	28	100	78	10	0.80	1.0
B1 five-shot ICL	67	70	60	100	11	3	0.79	1.0
B2 uniform memory	60	60	52	100	22	0	0.85	1.0
B3 fixed orchestration	60	63	52	100	11	7	0.63	2.0
control: blank query	0	17	0	0	100	100	0.00	1.0
control: permuted names	40	43	44	20	0	3	0.75	1.0
control: reversed options	60	63	52	100	22	3	0.83	1.0
DeepWeeds, final session 2, 30 known / 10 held-out queries								
B12 retrieve-then-verify	10	40	12	0	100	83	0.65	1.0
B13 one-vs-one yes/no	17	37	8	60	90	70	0.65	3.0
B0 zero-shot	17	27	20	0	50	17	0.85	1.0
B1 five-shot ICL	33	33	40	0	10	10	0.60	1.0
B2 uniform memory	30	43	28	40	60	37	0.73	1.0
B3 fixed orchestration	30	37	28	40	60	53	0.57	2.0
control: blank query	0	17	0	0	100	100	0.00	1.0
control: permuted names	13	23	16	0	80	50	0.73	1.0
control: reversed options	30	40	28	40	70	50	0.74	1.0
index-only top-1 (reference, not Qwen)	50	–	–	–	–	–	–	0

Table 6. Coverage versus decision (development, %).

Data set	t	$\|C_t\|$	Cov@1	Cov@3	B1	B2	B12	B2|cov	Attractor	Expected	Rev.
Mango Leaf	0	3	100	100	80	87	–	87	47	33	100
Mango Leaf	1	5	60	84	68	56	–	53	40	20	76
Mango Leaf	2	6	67	90	70	60	–	61	37	17	83
DeepWeeds	0	3	93	100	87	60	60	60	60	33	73
DeepWeeds	1	5	56	96	28	36	36	35	64	20	32
DeepWeeds	2	6	50	80	33	43	40	48	33	17	50

Diseases (7 classes) (class order 42; calibration thresholds from a separate known-class development draw). Under the pre-registered rule (open-world balanced accuracy, $\geq$ 3 points, interval above zero, $p_{\mathrm{Holm}} < 0.05$, on at least two data sets) the improvement over B2 is met on 1 of 10 data sets (IP102 (25 cl.)). Table 12 gives all values, Table 13 all contrasts.

Data sets with at least 25 classes. The only data set meeting the rule above is the largest one, IP102. After seeing this, and before running it, we pre-registered an extension (rounds/R5/hypotheses_25plus.md): dense memory should help when the 128 KiB budget is spread over many classes. It covers every locked manifest with at least 25 classes (IP102, iNatAg-25, iNatAg-100) and adds B12, the same retrieve-then-verify orchestration on the uniform memory, so that D12−B12 isolates the memory optimisation at equal orchestration. B2 and B1 cannot be run on iNatAg-100: 40–80 released classes exceed the single-token option alphabet of the decision rule. iNatAg-100 query images that appear in any iNatAg-25 cohort are excluded. Only iNatAg-100 and class orders 43/44 are confirmatory here; IP102 and iNatAg-25 order 42 generated the hypothesis. Contrasts meeting the rule (separate Holm family, order 42): IP102 (25 cl.) vs B2, PlantDoc vs B12.

Table 7. Pre-registered confirmation on never-examined image groups, final session (columns as in Table 5).

Arm	BA	BA$_f$	Old	New	Rej.	FR	AUROC	Calls
DeepWeeds, final session 2, 18 known / 6 held-out queries								
C1 indep. verification (CGV)	17	17	13	33	67	61	0.62	7.0
B12 retrieve-then-verify	6	39	7	0	100	89	0.69	1.0
B0 zero-shot	28	33	27	33	33	22	0.61	1.0
B1 five-shot ICL	33	39	40	0	17	17	0.58	1.0
B2 uniform memory	17	39	20	0	33	61	0.59	1.0
index-only top-1 (reference, not Qwen)	39	–	–	–	–	–	–	0
Dangerous Insects, final session 2, 33 known / 12 held-out queries								
C1 indep. verification (CGV)	67	70	79	33	42	12	0.71	12.0
B12 retrieve-then-verify	33	39	37	22	100	39	0.81	1.0
B0 zero-shot	67	67	75	44	33	0	0.85	1.0
B1 five-shot ICL	55	58	62	33	17	9	0.84	1.0
B2 uniform memory	61	64	67	44	50	9	0.85	1.0
index-only top-1 (reference, not Qwen)	33	–	–	–	–	–	–	0

Table 8. Pre-registered paired contrasts on the confirmatory cohorts (points). The confirmatory family is the open-world rows; forced-choice rows are secondary. Holm correction within each metric over 3 contrasts × 2 data sets.

Data set	Contrast	Metric	Δ	95% CI	p	p_{Holm}	McNemar p	Discordant	n
DeepWeeds	C1 − B2	open	+0.0	[-22.2, +16.7]	1.000	1.000	1.000	2/2	18
DeepWeeds	B12 − B2	open	-11.1	[-16.7, +0.0]	0.084	0.405	0.500	0/2	18
DeepWeeds	B1 − B2	open	+16.7	[+0.0, +33.3]	0.081	0.405	0.453	5/2	18
Dangerous Insects	C1 − B2	open	+6.1	[-6.1, +18.2]	0.426	0.852	0.754	6/4	33
Dangerous Insects	B12 − B2	open	-27.3	[-39.4, -15.2]	0.001	0.003	0.022	2/11	33
Dangerous Insects	B1 − B2	open	-6.1	[-15.2, +3.0]	0.274	0.822	0.625	1/3	33
DeepWeeds	C1 − B2	forced	-22.2	[-44.4, +0.0]	0.058	0.290	0.289	2/6	18
DeepWeeds	B12 − B2	forced	+0.0	[-16.7, +16.7]	1.000	1.000	1.000	3/3	18
DeepWeeds	B1 − B2	forced	+0.0	[-11.1, +11.1]	1.000	1.000	1.000	1/1	18
Dangerous Insects	C1 − B2	forced	+6.1	[-9.1, +18.2]	0.484	1.000	0.754	6/4	33
Dangerous Insects	B12 − B2	forced	-24.2	[-36.4, -12.1]	0.001	0.003	0.039	2/10	33
Dangerous Insects	B1 − B2	forced	-6.1	[-15.2, +3.0]	0.274	1.000	0.625	1/3	33

Table 9. Model calls, prompt tokens, images per prompt and seconds per query on the confirmatory cohorts.

Arm	Records	Calls	Tokens	Images	s (mean)	s (p95)
B0 zero-shot	185.0	1.0	364	1.0	2.0	2.4
B1 five-shot ICL	184.0	1.0	1338	14.7	5.5	8.3
B12 retrieve-then-verify	184.0	1.0	730	5.1	2.8	3.5
B2 uniform memory	184.0	1.0	1194	9.1	4.3	5.5
C1 indep. verification (CGV)	184.0	7.8	501	1.0	14.5	22.1

Table 10. Round 5 development (class order 42, final session; DeepWeeds and Mango three, Insects two queries per class). BA open / forced / calibrated (%; calibration in-sample on development, descriptive only); old/new; Rej. and Rej.$_{\text{cal}}$: held-out answered UNKNOWN; AUROC; calls.

Arm	BA	BA$_f$	BA$_{\text{cal}}$	Old	New	Rej.	Rej.$_{\text{cal}}$	AUROC	Calls
Mango Leaf									
B1 five-shot ICL (raw supports)	67	72	72	60	100	17	0	0.81	1.0
B2 uniform memory	67	67	67	60	100	17	0	0.83	1.0
D2 dense memory	56	61	56	47	100	17	0	0.76	1.0
D3 dense + fixed orch.	61	72	67	53	100	0	0	0.63	2.0
D4 dense + autonomous orch.	61	61	61	53	100	50	17	0.86	3.7
D12 dense + retrieve-then-verify	61	78	67	53	100	50	0	0.67	1.0
DeepWeeds									
B1 five-shot ICL (raw supports)	39	39	39	47	0	17	0	0.52	1.0
B2 uniform memory	28	44	39	27	33	67	17	0.69	1.0
D2 dense memory	22	39	33	27	0	50	33	0.62	1.0
D3 dense + fixed orch.	22	22	22	27	0	67	17	0.65	2.0
D4 dense + autonomous orch.	17	22	22	20	0	50	0	0.61	3.8
D12 dense + retrieve-then-verify	39	61	56	40	33	67	67	0.69	1.0
Dangerous Insects									
B1 five-shot ICL (raw supports)	55	55	55	69	17	62	62	0.95	1.0
B2 uniform memory	64	64	64	81	17	88	75	0.98	1.0
D2 dense memory	55	55	55	75	0	88	75	0.99	1.0
D3 dense + fixed orch.	64	73	68	75	33	88	88	0.98	2.0
D4 dense + autonomous orch.	73	73	73	81	50	75	88	0.98	3.3
D12 dense + retrieve-then-verify	59	68	64	69	33	88	88	0.98	1.0
IDC									
B1 five-shot ICL (raw supports)	25	25	25	0	100	—	—	—	1.0
B2 uniform memory	25	33	33	22	33	—	—	—	1.0
D12 dense + retrieve-then-verify	33	50	42	33	33	—	—	—	1.0

Table 11. Round 5 paired development contrasts (exploratory).

Data set	Contrast	Metric	Δ	95% CI	p	p_{Holm}	McNemar	Discordant	n
DeepWeeds	D2 − B2	open	-5.6	[-16.7, +0.0]	0.565	1.000	1.000	0/1	18
DeepWeeds	D3 − B2	open	-5.6	[-22.2, +11.1]	0.717	1.000	1.000	1/2	18
DeepWeeds	D4 − B2	open	-11.1	[-22.2, +0.0]	0.160	1.000	0.500	0/2	18
DeepWeeds	D12 − B2	open	+11.1	[-5.6, +27.8]	0.386	1.000	0.625	3/1	18
DeepWeeds	D2 − B1	open	-16.7	[-27.8, -5.6]	0.019	0.380	0.250	0/3	18
DeepWeeds	B1 − B2	open	+11.1	[-5.6, +27.8]	0.115	1.000	0.625	3/1	18
IDC	D12 − B2	open	+8.3	[-16.7, +33.3]	0.700	1.000	1.000	3/2	12
IDC	B1 − B2	open	+0.0	[-16.7, +16.7]	0.826	1.000	1.000	2/2	12
Dangerous Insects	D2 − B2	open	-9.1	[-18.2, +0.0]	0.116	1.000	0.500	0/2	22
Dangerous Insects	D3 − B2	open	+0.0	[-9.1, +9.1]	1.000	1.000	1.000	1/1	22
Dangerous Insects	D4 − B2	open	+9.1	[-4.5, +22.7]	0.385	1.000	0.688	4/2	22
Dangerous Insects	D12 − B2	open	-4.5	[-22.7, +9.1]	0.784	1.000	1.000	2/3	22
Dangerous Insects	D2 − B1	open	+0.0	[-9.1, +9.1]	1.000	1.000	1.000	1/1	22
Dangerous Insects	B1 − B2	open	-9.1	[-22.7, +4.5]	0.298	1.000	0.625	1/3	22
Mango Leaf	D2 − B2	open	-11.1	[-22.2, +0.0]	0.176	1.000	0.500	0/2	18
Mango Leaf	D3 − B2	open	-5.6	[-22.2, +11.1]	0.688	1.000	1.000	2/3	18
Mango Leaf	D4 − B2	open	-5.6	[-16.7, +0.0]	0.566	1.000	1.000	0/1	18
Mango Leaf	D12 − B2	open	-5.6	[-22.2, +16.7]	0.636	1.000	1.000	2/3	18
Mango Leaf	D2 − B1	open	-11.1	[-16.7, +0.0]	0.077	1.000	0.500	0/2	18
Mango Leaf	B1 − B2	open	+0.0	[-11.1, +11.1]	1.000	1.000	1.000	1/1	18
DeepWeeds	D2 − B2	forced	-5.6	[-16.7, +0.0]	0.565	1.000	1.000	0/1	18
DeepWeeds	D3 − B2	forced	-22.2	[-33.3, -16.7]	0.001	0.010	0.125	0/4	18
DeepWeeds	D4 − B2	forced	-22.2	[-33.3, -16.7]	0.001	0.010	0.125	0/4	18
DeepWeeds	D12 − B2	forced	+16.7	[-0.1, +33.3]	0.167	1.000	0.375	4/1	18
DeepWeeds	D2 − B1	forced	+0.0	[+0.0, +0.0]	1.000	1.000	1.000	0/0	18
DeepWeeds	B1 − B2	forced	-5.6	[-16.7, +0.0]	0.565	1.000	1.000	0/1	18
IDC	D12 − B2	forced	+16.7	[-8.3, +50.0]	0.489	1.000	0.688	4/2	12
IDC	B1 − B2	forced	-8.3	[-25.0, +8.3]	0.639	1.000	1.000	1/2	12
Dangerous Insects	D2 − B2	forced	-9.1	[-18.2, +0.0]	0.116	1.000	0.500	0/2	22
Dangerous Insects	D3 − B2	forced	+9.1	[+0.0, +18.2]	0.118	1.000	0.625	3/1	22
Dangerous Insects	D4 − B2	forced	+9.1	[-4.5, +22.7]	0.385	1.000	0.688	4/2	22
Dangerous Insects	D12 − B2	forced	+4.5	[-13.6, +18.2]	0.834	1.000	1.000	4/3	22
Dangerous Insects	D2 − B1	forced	+0.0	[-9.1, +9.1]	1.000	1.000	1.000	1/1	22
Dangerous Insects	B1 − B2	forced	-9.1	[-22.7, +4.5]	0.298	1.000	0.625	1/3	22
Mango Leaf	D2 − B2	forced	-5.6	[-16.7, +0.0]	0.566	1.000	1.000	0/1	18
Mango Leaf	D3 − B2	forced	+5.6	[-5.6, +16.7]	0.413	1.000	1.000	2/1	18
Mango Leaf	D4 − B2	forced	-5.6	[-16.7, +0.0]	0.566	1.000	1.000	0/1	18
Mango Leaf	D12 − B2	forced	+11.1	[-5.6, +27.8]	0.295	1.000	0.625	3/1	18
Mango Leaf	D2 − B1	forced	-11.1	[-16.7, +0.0]	0.077	1.000	0.500	0/2	18
Mango Leaf	B1 − B2	forced	+5.6	[+0.0, +16.7]	0.599	1.000	1.000	1/0	18
DeepWeeds	D2 − B2	calibrated	-5.6	[-16.7, +0.0]	0.565	1.000	1.000	0/1	18
DeepWeeds	D3 − B2	calibrated	-16.7	[-27.8, -5.6]	0.020	0.400	0.375	1/4	18
DeepWeeds	D4 − B2	calibrated	-16.7	[-27.8, -5.6]	0.020	0.400	0.375	1/4	18
DeepWeeds	D12 − B2	calibrated	+16.7	[+0.0, +33.3]	0.182	1.000	0.375	4/1	18
DeepWeeds	D2 − B1	calibrated	-5.6	[-16.7, +0.0]	0.602	1.000	1.000	0/1	18
DeepWeeds	B1 − B2	calibrated	+0.0	[-11.1, +11.1]	1.000	1.000	1.000	1/1	18
IDC	D12 − B2	calibrated	+8.3	[-25.0, +41.7]	0.826	1.000	1.000	4/3	12
IDC	B1 − B2	calibrated	-8.3	[-25.0, +8.3]	0.639	1.000	1.000	1/2	12
Dangerous Insects	D2 − B2	calibrated	-9.1	[-18.2, +0.0]	0.116	1.000	0.500	0/2	22
Dangerous Insects	D3 − B2	calibrated	+4.5	[+0.0, +9.1]	0.460	1.000	1.000	2/1	22
Dangerous Insects	D4 − B2	calibrated	+9.1	[-4.5, +22.7]	0.385	1.000	0.688	4/2	22
Dangerous Insects	D12 − B2	calibrated	+0.0	[-18.2, +18.2]	1.000	1.000	1.000	3/3	22
Dangerous Insects	D2 − B1	calibrated	+0.0	[-9.1, +9.1]	1.000	1.000	1.000	1/1	22
Dangerous Insects	B1 − B2	calibrated	-9.1	[-22.7, +4.5]	0.298	1.000	0.625	1/3	22
Mango Leaf	D2 − B2	calibrated	-11.1	[-22.2, +0.0]	0.176	1.000	0.500	0/2	18
Mango Leaf	D3 − B2	calibrated	+0.0	[-16.7, +16.7]	1.000	1.000	1.000	2/2	18
Mango Leaf	D4 − B2	calibrated	-5.6	[-16.7, +0.0]	0.566	1.000	1.000	0/1	18
Mango Leaf	D12 − B2	calibrated	+0.0	[-16.8, +16.8]	1.000	1.000	1.000	2/2	18
Mango Leaf	D2 − B1	calibrated	-16.7	[-27.8, -5.6]	0.028	0.504	0.250	0/3	18
Mango Leaf	B1 − B2	calibrated	+5.6	[+0.0, +16.7]	0.599	1.000	1.000	1/0	18

Table 12. Round 5 confirmation on never-examined groups (final session, mean over completed class orders).

Arm	BA	BA_f	$\mathrm{BA}_{\mathrm{cal}}$	Old	New	Rej.	$\mathrm{Rej.}_{\mathrm{cal}}$	AUROC	Calls
DeepWeeds									
B1 five-shot ICL (raw supports)	39	50	41	38	44	39	11	0.60	1.0
B2 uniform memory	41	44	38	40	44	39	28	0.64	1.0
D12 dense + retrieve-then-verify	26	48	36	27	22	72	56	0.71	1.0
Dangerous Insects									
B1 five-shot ICL (raw supports)	61	67	61	67	44	67	58	0.83	1.0
B2 uniform memory	65	68	65	67	61	75	75	0.87	1.0
D12 dense + retrieve-then-verify	55	58	58	58	44	92	92	0.83	1.0
SBRD									
B1 five-shot ICL (raw supports)	58	64	59	50	100	17	0	0.58	1.0
B2 uniform memory	53	53	50	53	50	0	0	0.52	1.0
D12 dense + retrieve-then-verify	44	58	52	40	67	67	39	0.77	1.0
Soybean Seeds									
B1 five-shot ICL (raw supports)	71	79	80	67	83	50	0	0.78	1.0
B2 uniform memory	71	71	77	61	100	0	0	0.62	1.0
D12 dense + retrieve-then-verify	58	67	60	50	83	33	33	0.78	1.0
Bean Leaf									
B1 five-shot ICL (raw supports)	56	56	56	58	50	—	—	—	1.0
B2 uniform memory	72	72	72	83	50	—	—	—	1.0
D12 dense + retrieve-then-verify	50	50	50	75	0	—	—	—	1.0
iNatAg-25									
B1 five-shot ICL (raw supports)	53	56	56	54	49	36	4	0.67	1.0
B2 uniform memory	50	52	53	56	31	44	4	0.70	1.0
D12 dense + retrieve-then-verify	56	62	64	56	53	58	7	0.67	1.0
IP102 (25 cl.)									
B1 five-shot ICL (raw supports)	25	26	26	24	29	49	18	0.66	1.0
B2 uniform memory	14	16	15	16	11	67	16	0.63	1.0
D12 dense + retrieve-then-verify	31	48	43	27	44	69	36	0.66	1.0
Soybean Diseases (7 cl.)									
B1 five-shot ICL (raw supports)	36	39	38	33	50	0	0	0.74	1.0
B2 uniform memory	33	33	33	30	50	0	0	0.89	1.0
D12 dense + retrieve-then-verify	39	50	43	30	83	17	0	0.54	1.0
PlantDoc									
D12 dense + retrieve-then-verify	47	53	52	44	58	20	7	0.56	1.0
B12 uniform + retrieve-then-verify	20	20	20	24	0	27	0	0.48	1.0
Yellow Rust 19									
B1 five-shot ICL (raw supports)	40	40	33	50	0	0	0	0.84	1.0
B2 uniform memory	47	47	40	58	0	0	0	0.82	1.0
D12 dense + retrieve-then-verify	33	33	27	33	33	100	33	1.00	1.0
FUSARIUM 22									
B1 five-shot ICL (raw supports)	50	50	42	44	67	0	0	0.58	1.0
B2 uniform memory	25	25	25	11	67	0	0	0.36	1.0
D12 dense + retrieve-then-verify	33	42	33	44	0	67	0	0.44	1.0

Table 13. Round 5 pre-registered contrasts, class order 42 (open: primary; forced and calibrated: secondary).

Data set	Contrast	Metric	Δ	95% CI	p	p_{Holm}	McNemar	Discordant	n
Bean Leaf	D12 − B2	open	-44.4	[-77.8, -11.1]	0.026	0.432	0.219	1/5	9
Bean Leaf	D12 − B1	open	-33.3	[-55.6, -11.1]	0.024	0.432	0.453	2/5	9
DeepWeeds	D12 − B2	open	-16.7	[-27.8, -5.6]	0.029	0.435	0.250	0/3	18
DeepWeeds	D12 − B1	open	-16.7	[-27.8, -5.6]	0.024	0.432	0.375	1/4	18
FUSARIUM 22	D12 − B2	open	+8.3	[-16.7, +33.3]	0.743	1.000	1.000	4/3	12
FUSARIUM 22	D12 − B1	open	-16.7	[-50.0, +16.7]	0.430	1.000	0.727	3/5	12
Dangerous Insects	D12 − B2	open	-10.6	[-21.2, -0.0]	0.036	0.504	0.375	1/4	30
Dangerous Insects	D12 − B1	open	-6.1	[-21.2, +9.1]	0.523	1.000	0.727	3/5	30
IP102 (25 cl.)	D12 − B2	open	+25.0	[+16.7, +33.3]	0.001	0.010	0.000	16/1	60
IP102 (25 cl.)	D12 − B1	open	+21.7	[+13.3, +31.7]	0.001	0.010	0.001	14/1	60
SBRD	D12 − B2	open	+0.0	[-16.7, +16.7]	0.962	1.000	1.000	3/3	18
SBRD	D12 − B1	open	-11.1	[-22.2, +0.0]	0.166	1.000	0.500	0/2	18
Soybean Diseases (7 cl.)	D12 − B2	open	-5.6	[-22.2, +11.1]	0.684	1.000	1.000	4/5	18
Soybean Diseases (7 cl.)	D12 − B1	open	-5.6	[-22.2, +11.1]	0.700	1.000	1.000	4/5	18
Soybean Seeds	D12 − B2	open	+0.0	[-16.7, +16.7]	1.000	1.000	1.000	1/1	12
Soybean Seeds	D12 − B1	open	-8.3	[-25.0, +0.0]	0.580	1.000	1.000	2/3	12
Yellow Rust 19	D12 − B2	open	-13.3	[-40.0, +13.3]	0.227	1.000	0.688	2/4	15
Yellow Rust 19	D12 − B1	open	-6.7	[-33.3, +13.3]	0.548	1.000	1.000	2/3	15
iNatAg-25	D12 − B2	open	+3.3	[-6.7, +13.3]	0.658	1.000	0.815	10/8	60
iNatAg-25	D12 − B1	open	+5.0	[-6.7, +16.7]	0.457	1.000	0.648	11/8	60
Bean Leaf	D12 − B2	forced	-44.4	[-77.8, -11.1]	0.026	0.416	0.219	1/5	9
Bean Leaf	D12 − B1	forced	-33.3	[-55.6, -11.1]	0.024	0.408	0.453	2/5	9
DeepWeeds	D12 − B2	forced	+5.6	[-11.1, +22.2]	0.760	1.000	1.000	2/1	18
DeepWeeds	D12 − B1	forced	+5.6	[-11.1, +22.2]	0.760	1.000	1.000	2/1	18
FUSARIUM 22	D12 − B2	forced	+16.7	[-8.3, +41.7]	0.290	1.000	0.688	4/2	12
FUSARIUM 22	D12 − B1	forced	-8.3	[-41.7, +25.0]	0.788	1.000	1.000	3/4	12
Dangerous Insects	D12 − B2	forced	-10.6	[-24.2, +1.5]	0.085	1.000	0.453	2/5	30
Dangerous Insects	D12 − B1	forced	-9.1	[-24.2, +3.0]	0.191	1.000	0.508	3/6	30
IP102 (25 cl.)	D12 − B2	forced	+40.0	[+30.0, +50.0]	0.001	0.010	0.000	25/1	60
IP102 (25 cl.)	D12 − B1	forced	+35.0	[+23.3, +46.7]	0.001	0.010	0.000	23/2	60
SBRD	D12 − B2	forced	+22.2	[+11.1, +33.3]	0.004	0.072	0.219	5/1	18
SBRD	D12 − B1	forced	+0.0	[-22.2, +16.7]	1.000	1.000	1.000	2/2	18
Soybean Diseases (7 cl.)	D12 − B2	forced	+5.6	[-11.1, +22.2]	0.364	1.000	1.000	6/5	18
Soybean Diseases (7 cl.)	D12 − B1	forced	+5.6	[-11.1, +22.2]	0.383	1.000	1.000	6/5	18
Soybean Seeds	D12 − B2	forced	+8.3	[+0.0, +25.0]	0.601	1.000	1.000	1/0	12
Soybean Seeds	D12 − B1	forced	-8.3	[-25.0, +8.3]	0.664	1.000	1.000	2/3	12
Yellow Rust 19	D12 − B2	forced	-13.3	[-40.0, +13.3]	0.227	1.000	0.688	2/4	15
Yellow Rust 19	D12 − B1	forced	-6.7	[-33.3, +13.3]	0.548	1.000	1.000	2/3	15
iNatAg-25	D12 − B2	forced	+10.0	[+1.7, +18.3]	0.032	0.480	0.263	13/7	60
iNatAg-25	D12 − B1	forced	+8.3	[-0.0, +18.3]	0.099	1.000	0.359	12/7	60
Bean Leaf	D12 − B2	calibrated	-44.4	[-77.8, -11.1]	0.026	0.416	0.219	1/5	9
Bean Leaf	D12 − B1	calibrated	-33.3	[-55.6, -11.1]	0.024	0.408	0.453	2/5	9
DeepWeeds	D12 − B2	calibrated	+0.0	[-16.7, +22.2]	1.000	1.000	1.000	2/2	18
DeepWeeds	D12 − B1	calibrated	+0.0	[-16.7, +22.2]	1.000	1.000	1.000	2/2	18
FUSARIUM 22	D12 − B2	calibrated	+8.3	[-16.7, +33.3]	0.743	1.000	1.000	4/3	12
FUSARIUM 22	D12 − B1	calibrated	-8.3	[-33.3, +16.7]	0.717	1.000	1.000	3/4	12
Dangerous Insects	D12 − B2	calibrated	-7.6	[-18.2, +3.0]	0.182	1.000	0.688	2/4	30
Dangerous Insects	D12 − B1	calibrated	-3.0	[-18.2, +12.1]	0.805	1.000	1.000	4/5	30
IP102 (25 cl.)	D12 − B2	calibrated	+36.7	[+26.7, +46.7]	0.001	0.010	0.000	23/1	60
IP102 (25 cl.)	D12 − B1	calibrated	+31.7	[+20.0, +43.3]	0.001	0.010	0.000	21/2	60
SBRD	D12 − B2	calibrated	+11.1	[-5.6, +27.8]	0.262	1.000	0.688	4/2	18
SBRD	D12 − B1	calibrated	-11.1	[-27.8, +5.6]	0.310	1.000	0.625	1/3	18
Soybean Diseases (7 cl.)	D12 − B2	calibrated	+5.6	[-11.1, +22.2]	0.364	1.000	1.000	6/5	18
Soybean Diseases (7 cl.)	D12 − B1	calibrated	+5.6	[-11.1, +22.2]	0.383	1.000	1.000	6/5	18
Soybean Seeds	D12 − B2	calibrated	-8.3	[-25.0, +0.0]	0.605	1.000	1.000	0/1	12
Soybean Seeds	D12 − B1	calibrated	-25.0	[-41.7, -8.3]	0.022	0.396	0.375	1/4	12
Yellow Rust 19	D12 − B2	calibrated	-13.3	[-40.0, +13.3]	0.318	1.000	0.688	2/4	15
Yellow Rust 19	D12 − B1	calibrated	-6.7	[-33.3, +13.3]	0.548	1.000	1.000	2/3	15
iNatAg-25	D12 − B2	calibrated	+10.0	[+1.7, +18.3]	0.032	0.480	0.263	13/7	60
iNatAg-25	D12 − B1	calibrated	+10.0	[+1.7, +18.3]	0.035	0.490	0.238	12/6	60

Table 14. Round 5 extension to data sets with $\geq$25 classes: D12 vs B12 (memory optimisation at equal orchestration) and vs B2, class order 42.

Data set	Contrast	Metric	Δ	95% CI	p	p_{Holm}	McNemar	Discordant	n
IP102 (25 cl.)	D12 − B2	open	+25.0	[+16.7, +33.3]	0.001	0.002	0.000	16/1	60
iNatAg-25	D12 − B2	open	+3.3	[-6.7, +13.3]	0.658	0.658	0.815	10/8	60
PlantDoc	D12 − B12	open	+27.3	[+18.2, +36.4]	0.001	0.002	0.000	21/3	66
IP102 (25 cl.)	D12 − B2	forced	+40.0	[+30.0, +50.0]	0.001	0.002	0.000	25/1	60
iNatAg-25	D12 − B2	forced	+10.0	[+1.7, +18.3]	0.032	0.032	0.263	13/7	60
PlantDoc	D12 − B12	forced	+33.3	[+25.8, +40.9]	0.001	0.002	0.000	23/1	66
IP102 (25 cl.)	D12 − B2	calibrated	+36.7	[+26.7, +46.7]	0.001	0.002	0.000	23/1	60
iNatAg-25	D12 − B2	calibrated	+10.0	[+1.7, +18.3]	0.032	0.032	0.263	13/7	60
PlantDoc	D12 − B12	calibrated	+31.8	[+24.2, +40.9]	0.001	0.002	0.000	22/1	66

Table 15. As Table 14, pooled over all completed class orders (orders crossed in the bootstrap).

Data set	Contrast	Metric	Δ	95% CI	p	p_{Holm}	McNemar	Discordant	n
IP102 (25 cl.)	D12 − B2	open	+16.7	[+8.3, +26.1]	0.001	0.002	0.000	37/7	180
iNatAg-25	D12 − B2	open	+5.6	[-1.1, +13.3]	0.152	0.152	0.229	33/23	180
PlantDoc	D12 − B12	open	+27.3	[+18.2, +36.4]	0.001	0.002	0.000	21/3	66
IP102 (25 cl.)	D12 − B2	forced	+32.2	[+23.3, +41.7]	0.001	0.002	0.000	66/8	180
iNatAg-25	D12 − B2	forced	+10.0	[+5.0, +15.6]	0.001	0.002	0.025	38/20	180
PlantDoc	D12 − B12	forced	+33.3	[+25.8, +40.9]	0.001	0.002	0.000	23/1	66
IP102 (25 cl.)	D12 − B2	calibrated	+26.7	[+17.2, +37.8]	0.001	0.002	0.000	56/8	180
iNatAg-25	D12 − B2	calibrated	+10.0	[+5.0, +15.6]	0.001	0.002	0.025	38/20	180
PlantDoc	D12 − B12	calibrated	+31.8	[+24.2, +40.9]	0.001	0.002	0.000	22/1	66

A. Complete numeric record

Every completed combination of cohort, data set, class order, session and arm, including arms that failed, generated from the prediction logs (results/per_session_*.csv). BA: open-world balanced accuracy (%); BA_f: forced choice; F1: open-world macro-F1; Old/New; Rej.: held-out answered UNKNOWN; FR: known answered UNKNOWN; AUROC; Qwen calls and seconds per query.

Cohort	Data set	Ord.	t	Arm	n_k	n_h	BA	BA_f	F1	Old	New	Rej.	FR	AUROC	Calls	s
Agent-dev	deepweeds	42	0	B0	15	10	13	53	19	–	13	100	60	0.94	1.0	1.4
Agent-dev	deepweeds	42	0	B1	15	10	80	87	85	–	80	40	13	0.80	1.0	2.9
Agent-dev	deepweeds	42	0	B12	15	10	60	60	61	–	60	30	13	0.77	1.0	3.2
Agent-dev	deepweeds	42	0	B13	15	10	40	67	50	–	40	70	60	0.59	3.0	5.8
Agent-dev	deepweeds	42	0	B2	15	10	60	60	61	–	60	30	13	0.77	1.0	3.1
Agent-dev	deepweeds	42	0	B2null	15	10	0	33	0	–	0	100	100	0.00	1.0	3.1
Agent-dev	deepweeds	42	0	B2perm	15	10	20	40	17	–	20	80	27	0.75	1.0	3.1
Agent-dev	deepweeds	42	0	B2rev	15	10	60	67	61	–	60	30	13	0.77	1.0	3.1
Agent-dev	deepweeds	42	0	B2x3	8	0	40	50	44	–	40	–	12	–	3.0	9.4
Agent-dev	deepweeds	42	0	B3	15	10	33	60	36	–	33	50	33	0.65	2.0	4.6
Agent-dev	deepweeds	42	0	B4	7	0	30	40	38	–	30	–	29	–	4.0	9.4
Agent-dev	deepweeds	42	1	B0	25	10	28	36	30	33	20	80	16	0.90	1.0	1.4
Agent-dev	deepweeds	42	1	B1	25	10	24	28	21	40	0	30	8	0.70	1.0	4.1
Agent-dev	deepweeds	42	1	B12	25	10	0	36	0	0	0	90	80	0.64	1.0	2.2
Agent-dev	deepweeds	42	1	B13	25	10	8	40	13	13	0	90	88	0.58	3.0	5.0
Agent-dev	deepweeds	42	1	B2	25	10	32	36	28	53	0	70	36	0.63	1.0	3.0
Agent-dev	deepweeds	42	1	B2null	25	10	0	20	0	0	0	100	100	0.00	1.0	2.9
Agent-dev	deepweeds	42	1	B2perm	25	10	16	24	14	13	20	60	32	0.65	1.0	3.0
Agent-dev	deepweeds	42	1	B2rev	25	10	24	28	21	20	30	40	20	0.63	1.0	3.0
Agent-dev	deepweeds	42	1	B3	25	10	20	44	20	33	0	80	60	0.69	2.0	3.9
Agent-dev	deepweeds	42	2	B0	30	10	17	27	21	20	0	50	17	0.85	1.0	1.4
Agent-dev	deepweeds	42	2	B1	30	10	33	33	24	40	0	10	10	0.60	1.0	4.7
Agent-dev	deepweeds	42	2	B12	30	10	10	40	14	12	0	100	83	0.65	1.0	2.3
Agent-dev	deepweeds	42	2	B13	30	10	17	37	20	8	60	90	70	0.65	3.0	5.0
Agent-dev	deepweeds	42	2	B2	30	10	30	43	30	28	40	60	37	0.73	1.0	3.0
Agent-dev	deepweeds	42	2	B2null	30	10	0	17	0	0	0	100	100	0.00	1.0	3.0
Agent-dev	deepweeds	42	2	B2perm	30	10	13	23	12	16	0	80	50	0.73	1.0	3.0
Agent-dev	deepweeds	42	2	B2rev	30	10	30	40	34	28	40	70	50	0.74	1.0	3.0
Agent-dev	deepweeds	42	2	B3	30	10	30	37	31	28	40	60	53	0.57	2.0	3.8
Agent-dev	mango	42	0	B0	15	9	33	60	33	–	33	100	60	0.70	1.0	1.5
Agent-dev	mango	42	0	B1	15	9	60	80	69	–	60	44	40	0.64	1.0	3.2
Agent-dev	mango	42	0	B2	15	9	80	87	83	–	80	0	13	0.53	1.0	5.1
Agent-dev	mango	42	0	B2null	15	9	0	33	0	–	0	100	100	0.00	1.0	5.1
Agent-dev	mango	42	0	B2perm	15	9	13	27	11	–	13	11	20	0.64	1.0	5.2
Agent-dev	mango	42	0	B2rev	15	9	80	87	83	–	80	0	13	0.58	1.0	5.2
Agent-dev	mango	42	0	B3	15	9	67	87	75	–	67	0	27	0.56	2.0	6.7
Agent-dev	mango	42	0	C1	1	0	0	0	0	–	0	–	100	–	4.0	9.1
Agent-dev	mango	42	1	B0	25	9	36	52	31	33	40	78	32	0.71	1.0	1.5
Agent-dev	mango	42	1	B1	25	9	64	68	68	73	50	11	16	0.47	1.0	4.4
Agent-dev	mango	42	1	B2	25	9	52	56	53	60	40	0	8	0.62	1.0	5.1
Agent-dev	mango	42	1	B2null	25	9	0	20	0	0	0	100	100	0.00	1.0	5.1
Agent-dev	mango	42	1	B2perm	25	9	44	44	35	53	30	0	4	0.68	1.0	4.6
Agent-dev	mango	42	1	B2rev	25	9	44	48	45	47	40	0	12	0.57	1.0	5.1
Agent-dev	mango	42	1	B3	25	9	52	72	54	73	20	44	28	0.54	2.0	5.9
Agent-dev	mango	42	2	B0	30	9	40	40	32	28	100	78	10	0.80	1.0	1.5
Agent-dev	mango	42	2	B1	30	9	67	70	59	60	100	11	3	0.79	1.0	5.0
Agent-dev	mango	42	2	B2	30	9	60	60	52	52	100	22	0	0.85	1.0	5.1
Agent-dev	mango	42	2	B2null	30	9	0	17	0	0	0	100	100	0.00	1.0	5.1
Agent-dev	mango	42	2	B2perm	30	9	40	43	35	44	20	0	3	0.75	1.0	5.1
Agent-dev	mango	42	2	B2rev	30	9	60	63	54	52	100	22	3	0.83	1.0	5.1
Agent-dev	mango	42	2	B3	30	9	60	63	56	52	100	11	7	0.63	2.0	5.3
Agent-fresh	deepweeds	42	0	B0	9	6	0	56	0	–	0	83	78	0.63	1.0	2.1
Agent-fresh	deepweeds	42	0	B1	9	6	56	67	59	–	56	33	22	0.57	1.0	2.9
Agent-fresh	deepweeds	42	0	B12	9	6	56	78	62	–	56	17	33	0.59	1.0	3.2
Agent-fresh	deepweeds	42	0	B2	9	6	56	78	62	–	56	17	33	0.59	1.0	3.2
Agent-fresh	deepweeds	42	0	C1	9	6	0	56	0	–	0	83	89	0.35	4.0	7.3
Agent-fresh	deepweeds	42	1	B0	15	6	27	40	23	33	17	67	20	0.70	1.0	2.1
Agent-fresh	deepweeds	42	1	B1	15	6	33	40	30	22	50	17	33	0.72	1.0	4.1
Agent-fresh	deepweeds	42	1	B12	15	6	7	33	8	11	0	100	87	0.63	1.0	2.2
Agent-fresh	deepweeds	42	1	B2	15	6	13	27	10	22	0	50	60	0.71	1.0	3.0
Agent-fresh	deepweeds	42	1	C1	15	6	13	27	13	22	0	100	80	0.62	6.0	9.8
Agent-fresh	deepweeds	42	2	B0	18	6	28	33	31	27	33	33	22	0.61	1.0	1.5
Agent-fresh	deepweeds	42	2	B1	18	6	33	39	28	40	0	17	17	0.58	1.0	4.7
Agent-fresh	deepweeds	42	2	B12	18	6	6	39	8	7	0	100	89	0.69	1.0	2.3
Agent-fresh	deepweeds	42	2	B2	18	6	17	39	15	20	0	33	61	0.59	1.0	3.0
Agent-fresh	deepweeds	42	2	C1	18	6	17	17	16	13	33	67	61	0.62	7.0	11.3
Agent-fresh	insects	42	0	B0	15	12	73	73	71	–	73	58	0	0.92	1.0	2.4
Agent-fresh	insects	42	0	B1	15	12	60	60	56	–	60	42	0	0.93	1.0	4.4
Agent-fresh	insects	42	0	B12	15	12	80	80	78	–	80	92	0	0.99	1.0	3.5

Cohort	Data set	Ord.	t	Arm	n_k	n_h	BA	BA$_f$	F1	Old	New	Rej.	FR	AUROC	Calls	s
Agent-fresh	insects	42	0	B2	15	12	73	73	70	–	73	58	0	0.93	1.0	4.9
Agent-fresh	insects	42	0	C1	15	12	80	87	83	–	80	25	7	0.81	6.0	12.5
Agent-fresh	insects	42	1	B0	24	12	79	79	77	80	78	17	0	0.91	1.0	2.2
Agent-fresh	insects	42	1	B1	24	12	67	67	65	60	78	8	4	0.88	1.0	6.7
Agent-fresh	insects	42	1	B12	24	12	50	58	51	33	78	100	38	0.86	1.0	2.8
Agent-fresh	insects	42	1	B2	24	12	67	67	67	60	78	50	12	0.86	1.0	5.1
Agent-fresh	insects	42	1	C1	24	12	83	83	84	80	89	25	4	0.76	9.0	17.4
Agent-fresh	insects	42	2	B0	33	12	67	67	59	75	44	33	0	0.85	1.0	1.8
Agent-fresh	insects	42	2	B1	33	12	55	58	55	62	33	17	9	0.84	1.0	8.3
Agent-fresh	insects	42	2	B12	33	12	33	39	32	37	22	100	39	0.81	1.0	2.5
Agent-fresh	insects	42	2	B2	33	12	61	64	59	67	44	50	9	0.85	1.0	5.4
Agent-fresh	insects	42	2	C1	33	12	67	70	67	79	33	42	12	0.71	12.0	22.0